

Real-Data Machine Learning for Telemedicine Triage and Outbreak Surveillance

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Real public data only · no synthetic records · seed-42 byte-identical reruns · fail-closed pipeline

Presentation Outline

- Introduction
- Problem Statement
- Methodology
- Results
- Conclusion
- Future Work
- References

Abstract

- Telemedicine platforms face two simultaneous decisions: which visit needs urgent escalation, and is a district-level outbreak beginning.
- Real public data only — CDC NHAMCS 2019 ED ($n = 13,595$), COVID-19 India district-weeks (126,712), Kaggle pharma sales; fail-closed if raw data is missing.
- Primary model fixed by a pre-declared rule (min OOF Brier): unweighted XGBoost + isotonic — AUC 0.742 [0.728, 0.751], Brier 0.113 vs 0.129 constant.
- Nested cross-fitted threshold: recall 0.829, under-triage 0.172, FPR 0.545.
- PAC split-conformal rule ($\delta = 0.10$): mean recall 0.900 with 3.5% of splits below target (marginal: 0.884 / 12.5%).
- Outbreak module: $7.0\times$ descriptive enrichment at a 5% alert budget — reported as enrichment, never as detection rates.

Introduction

- Two everyday decisions on a telemedicine platform, answered only with public data.
- Decision 1 — patient level: route an encounter to urgent care or routine advice. Errors are asymmetric: under-triage can be fatal.
- Decision 2 — population level: detect district-week outbreak signals under a fixed alert budget.
- Supplementary: pharmaceutical demand forecasting as a robustness check (naive baselines win 4/8 drugs — reported).
- Safety-first framing: sensitivity is prioritized, and every claim is audit-locked to a committed artifact.

Problem Statement

- Default threshold $t = 0.50$ is clinically unsafe: recall 0.141 $\rightarrow$ under-triage 85.9% (FPR 0.016).
- Thresholds tuned and evaluated on the same data overstate performance — nested cross-fitting is required for an honest estimate.
- The textbook conformal guarantee is only marginal: 12.5% of splits miss the 0.85 target.
- Outbreak data has no gold-standard labels — precision, recall and ROC cannot be honestly computed.
- Objective: conservative operating points + calibrated claims, all from real public data.

Literature Survey 1

Clinical severity scores (NEWS2):

- Vitals-only bedside score — cutoff ≥ 5 : recall 0.138, precision 0.207; best cutoff 1: 0.669 / 0.168.
- No cutoff ≥ 1 reaches 0.85 recall on calibration — matched comparison infeasible (documented).

Learned triage models:

- Models over reason-for-visit + vitals outperform vitals-only scoring at comparable precision.
- NHAMCS is the standard public ED benchmark for triage research [1].
- Limitation: US data — non-transferability to deployment settings is stated explicitly.

Literature Survey 2

Conformal prediction:

- Split-conformal prediction gives distribution-free marginal coverage [11], [12].
- Marginal $\neq$ per-split: the guarantee holds in expectation over calibration draws.

Risk control:

- PAC (probably-approximately-correct) guarantees bound the share of bad splits via β order statistics [11].

Our contribution:

- Apply PAC rank selection to triage recall control on a calibrated primary model
— marginal and PAC rules both reported over 200 seeds.

Literature Survey 3

Outbreak surveillance without labels:

- Open volunteer district archives (e.g. OpenDengue [13]) are rich in time series, poor in ground-truth labels.
- Standard practice: alert-budget concentration in verified wave windows (EARS-style) instead of ROC curves.

Forecasting baselines:

- Seasonal naive and Holt-Winters are strong competitors on low-variance series — the free-lunch finding is reported, not hidden [7].
- Limitation: descriptive enrichment only; retrospective on a static archive.

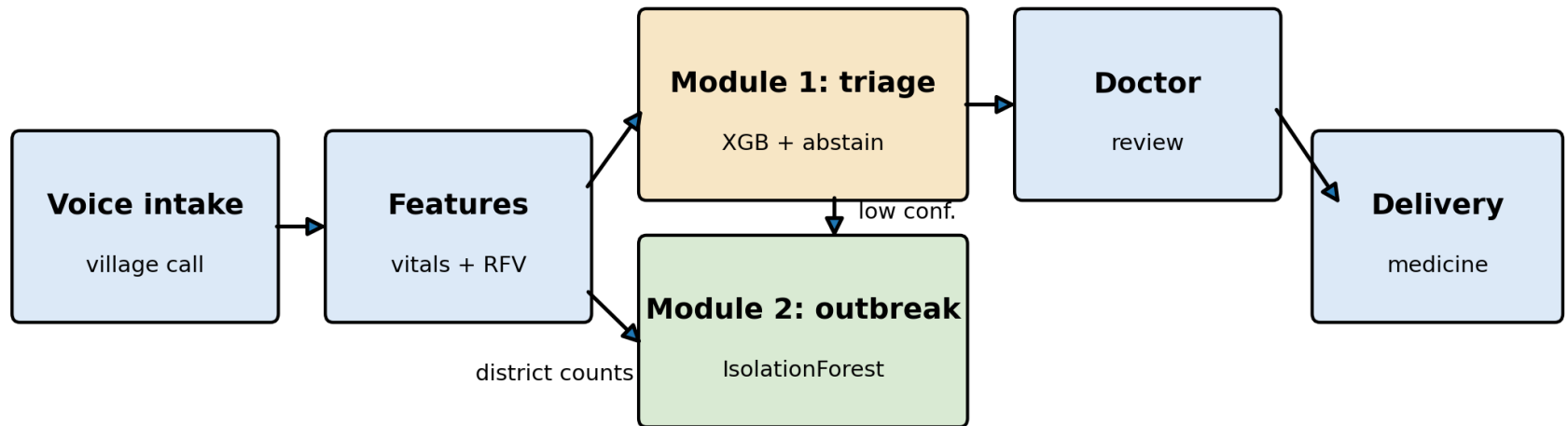
Proposed Methodology

- Six triage pipelines: LogReg, class-weighted XGB, unweighted XGB, + isotonic, + Platt, constant baseline.
- Primary chosen by pre-declared rule: minimum 5-fold OOF Brier among calibrated candidates (ties $\rightarrow$ isotonic).
- Threshold rule: argmax precision subject to OOF recall ≥ 0.80 (0.01 grid; ties $\rightarrow$ higher threshold).
- Nested cross-fitting: each outer fold re-selects its threshold on the other four folds only.
- Abstention band: calibrated $p \in (0.4, 0.6) \rightarrow$ human review (4.2% of cases).

Workflow: real data $\rightarrow$ calibrated model $\rightarrow$ cross-fitted threshold $\rightarrow$ PAC recall control $\rightarrow$ triage advice

Flowchart of Proposed Method

Curago data-science layer: patient-level triage + population-level surveillance



Data: CDC NHAMCS 2019 (public domain) | covid19india archive (static) | Kaggle pharma sales (CC BY-NC, suppl.)

Curago data-science layer: patient-level triage + population-level surveillance (Fig. 1 of the paper). Data: NHAMCS 2019 ED, inCOVID19 district archive, Kaggle pharma sales.

Primary Model & Calibration

- Primary: unweighted XGBoost + isotonic calibration — AUC 0.742 [0.728, 0.751].
- Logistic regression: AUC 0.719 [0.708, 0.731]; paired difference CI [0.012, 0.031] excludes zero.
- Class weighting distorts probabilities: Brier 0.165 vs 0.113 (isotonic).
- Calibrated model beats the constant baseline: $0.113 < 0.129$.
- Probabilities drive a threshold rule — they are not quoted as absolute risks.

Threshold Selection & Cross-Fitting

- Rule: argmax precision subject to OOF recall ≥ 0.80 (0.01 grid; ties $\rightarrow$ higher threshold).
- Pooled: $t = 0.11 \rightarrow$ recall 0.800, FPR 0.493, under-triage 0.200.
- Nested (headline): outer 5-fold — threshold re-tuned on the other four folds only, then evaluated on its own test fold.
- Nested result: $t = 0.09\text{--}0.10 \rightarrow$ recall 0.829, FPR 0.545, under-triage 0.172, abstention 0.042.
- No test-fold peeking — the conservative operating point we report.

PAC Split-Conformal Guarantee

- Protocol: 200 seeds, 60/20/20 stratified splits, $\alpha = 0.15$ (target recall 0.85).
- Marginal rule (rank $k = \lfloor \alpha(n+1) \rfloor$): mean 0.884 (SD 0.033), 12.5% of splits below target.
- PAC rule: largest k with $\text{BetaCDF}(1-\alpha; n+1-k, k) \leq \delta$, $\delta = 0.10$.
- PAC result: mean 0.900 (SD 0.030), 3.5% below target ($\leq \delta$), precision 0.191.
- PAC bounds the share of below-target splits — still not a per-split guarantee, and we say so.

Datasets Used

- CDC NHAMCS 2019 ED — 13,595 encounters after screening (US public domain).
- COVID-19 India district time-series — 126,712 district-weeks, 840 districts (open archive).
- Kaggle pharma sales — 69 monthly points, 8 ATC groups (CC BY-NC 4.0).
- Evaluation: 5-fold OOF CV, nested cross-fitting, 200-seed conformal study, 12-step forecast holdout.
- Every CSV carries a Data Source column; licenses and citations in docs/citations.md.

Results & Discussion

- Triage: under-triage 85.9% $\rightarrow$ 17.2% via cross-fitted thresholds (recall 0.829, FPR 0.545).
- Calibration: Brier 0.113 beats constant 0.129; probabilities are threshold inputs, not quoted risks.
- Guarantees: marginal 0.884 / 12.5% below target; PAC 0.900 / 3.5% ($\delta = 10\%$).
- Seed-42 split: recall 0.886 [0.856, 0.914], precision 0.199, FPR 0.643.
- Outbreak: $7.0\times$ enrichment (0.250 vs 0.036), 39.8% district coverage — descriptive.

Table 1. Triage model comparison (5-fold OOF, $t = 0.50$)

Model	AUC	Recall	Precision	Brier
Logistic regression	0.719	0.609	0.269	0.209
XGBoost (class-weighted)	0.732	0.513	0.334	0.165
XGBoost + isotonic (primary)	0.742	0.141	0.606	0.113
Constant (train prevalence)	0.500	0.000	0.000	0.129

Primary chosen by min OOF Brier among calibrated candidates. Data: CDC NHAMCS 2019 ED ($n = 13,595$).

Table 2. Operating points

Setting	t	Recall	Precision	FPR	Under-triage
Default, primary	0.50	0.141	0.605	0.016	0.859
Tuned, pooled OOF	0.11	0.800	0.226	0.493	0.200
Nested cross-fitted (headline)	0.09–0.10	0.829	0.214	0.545	0.172
Logistic regression, tuned	0.400	0.811	0.214	0.536	0.189
Conformal, seed-42 split	0.099	0.886	0.199	0.643	0.114

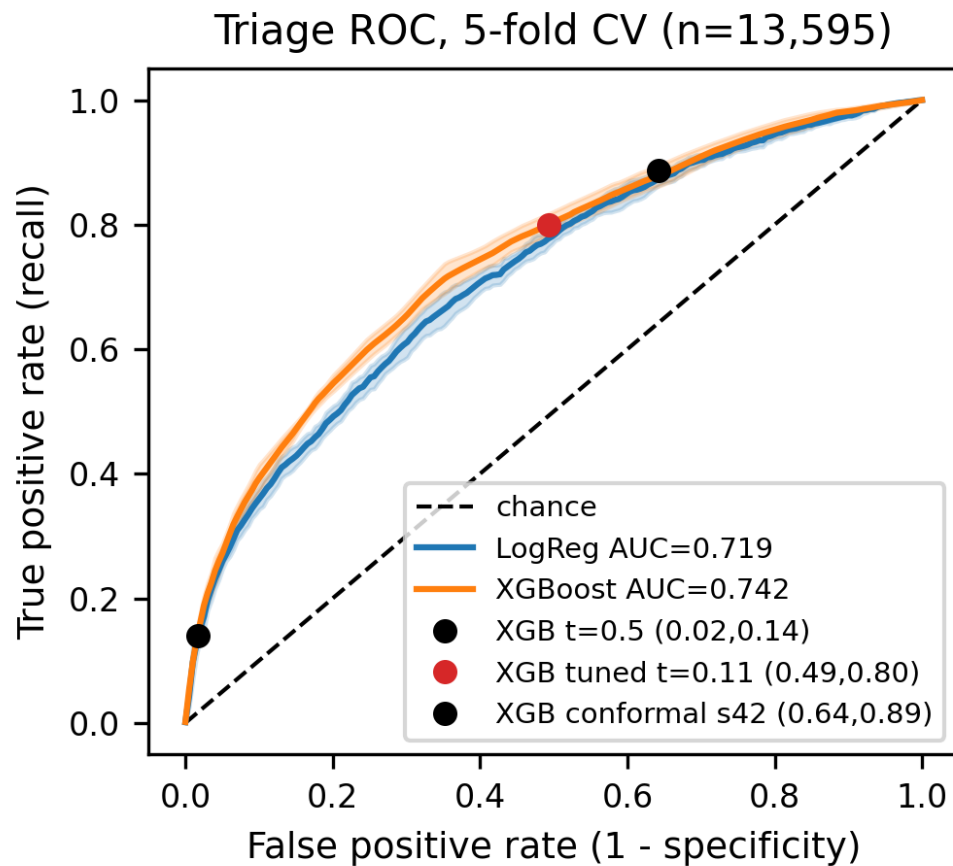
The nested row is the conservative headline: thresholds selected without test-fold access. Data: CDC NHAMCS 2019 ED.

Table 3. Conformal α sweep — marginal vs PAC (200 splits)

α	Rule	Mean recall	SD	Below target	Mean precision
0.05	marginal	0.966	0.015	14.5%	0.169
0.05	PAC ($\delta = 0.10$)	0.976	0.013	3.0%	0.164
0.10	marginal	0.924	0.023	15.5%	0.183
0.10	PAC ($\delta = 0.10$)	0.938	0.021	3.0%	0.179
0.15	marginal	0.884	0.033	12.5%	0.196
0.15	PAC ($\delta = 0.10$)	0.900	0.030	3.5%	0.191
0.20	marginal	0.832	0.038	22.0%	0.212
0.20	PAC ($\delta = 0.10$)	0.858	0.038	6.0%	0.204
0.30	marginal	0.737	0.043	17.5%	0.243
0.30	PAC ($\delta = 0.10$)	0.768	0.043	2.5%	0.233

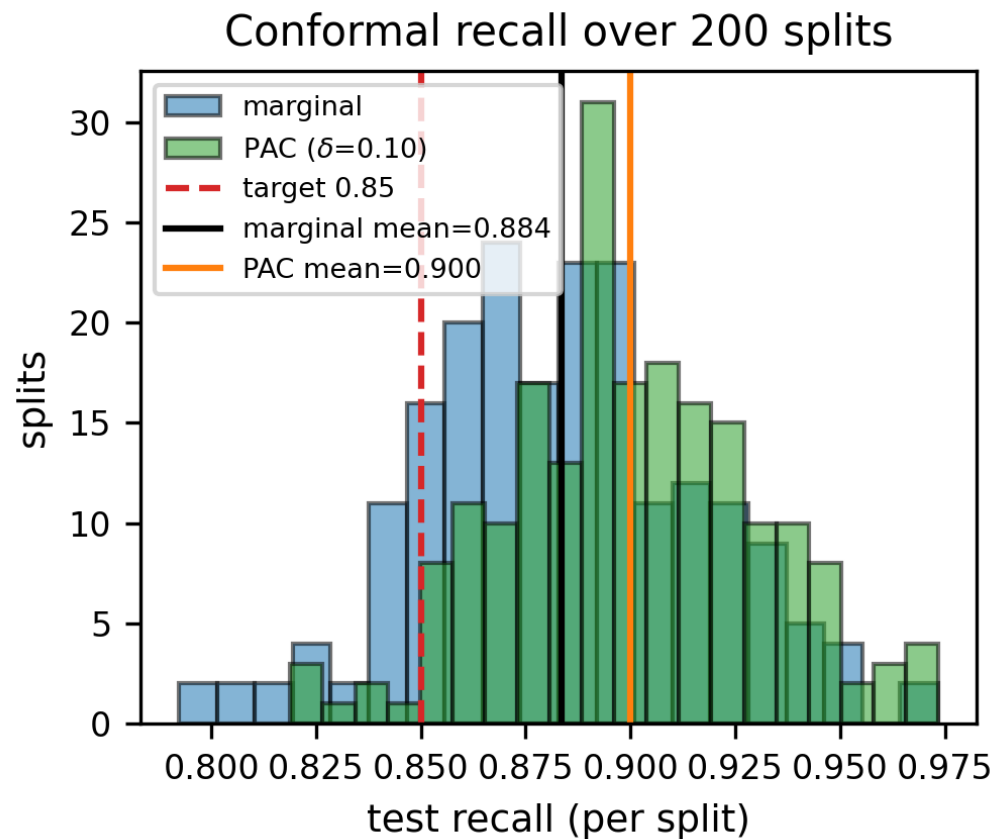
$\alpha = 0.15$ rows (bold) are the paper's headline: marginal 12.5% below target vs PAC $3.5\% \leq \delta$. Data: CDC NHAMCS 2019 ED.

ROC Curve



5-fold OOF ROC: primary AUC 0.742 [0.728, 0.751] vs LogReg 0.719 (Fig. 2 of the paper).

Conformal Recall Distribution



Test recall over 200 splits at $\alpha = 0.15$: marginal vs PAC rule (Fig. 5 of the paper).

Conclusion

- Real public data, no synthetic records — full pipeline fail-closed and seed-42 reproducible.
- Calibrated primary + nested cross-fitting: recall 0.829 at under-triage 0.172 (FPR 0.545).
- PAC conformal control: below-target share $3.5\% \leq \delta = 10\%$ (marginal 12.5%).
- Outbreak module: descriptive $7.0\times$ enrichment only — stated as such.

Future Work:

- External validation under covariate shift; outcome-linked labels (T4); surveillance baselines (T5); prospective live evaluation.

References

- [1] National Center for Health Statistics, "National hospital ambulatory medical care survey: 2019 emergency department summary," Centers for Disease Control and Prevention (CDC), 2019, data documentation: https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Dataset_Documentation/NHAMCS/doc19-ed-508.pdf. [Online]. Available: <https://www.cdc.gov/nchs/nhamcs/documentation/about-the-data-2019.html>
- [2] covid19india.com Data Operations Group and incovid19 Team, "Covid-19 india national and district-level time-series archive," Static data archive hosted at <https://data.incovid19.org/>, 2023, retrieved 2026-10-07. Retrospective observed series from 2020-04-26 to 2023-08-22 across 840 state-qualified districts.
- [3] M. Zdravković, "Pharma sales data: Point-of-sale transactional dataset resampled to monthly aggregates," Kaggle, 2019, license: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). [Online]. Available: <https://www.kaggle.com/datasets/milanzdravkovic/pharma-sales-data>
- [4] Royal College of Physicians, "National Early Warning Score (NEWS) 2: Standardising the Assessment of Acute-Illness Severity in the NHS". London, UK: Royal College of Physicians, 2017, report of a working party.
- [5] G. B. Smith, D. R. Prytherch, P. Meredith, P. E. Schmidt, and P. I. Featherstone, "The ability of the National Early Warning Score (NEWS) to discriminate patients at risk of early death in acute medical admissions," **Resuscitation**, vol. 84, no. 4, pp. 465–470, 2013.
- [6] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in **Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)**, 2016, pp. 785–794.
- [7] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in **Advances in Neural Information Processing Systems (NeurIPS 2017)**, vol. 30, 2017, pp. 3146–3154.
- [8] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," **Journal of Machine Learning Research (JMLR)**, vol. 12, pp. 2825–2830, 2011.
- [9] V. Vovk, A. Gammerman, and G. Shafer, **Algorithmic Learning in a Random World**. New York, NY, USA: Springer Science & Business Media, 2005.
- [10] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," **arXiv preprint arXiv:2107.07511**, 2021.
- [11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in **Advances in Neural Information Processing Systems (NeurIPS 2017)**, vol. 30, 2017, pp. 4765–4774.
- [12] J. Clarke, M. U. G. Kraemer, and et al., "OpenDengue: Data platform for global dengue surveillance," **Scientific Data**, vol. 11, no. 1, p. 324, 2024.
- [13] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in **Proceedings of the 9th Python in Science Conference (SciPy 2010)**, 2010, pp. 92–96.

Thank You

Questions?

*Paper: Real-Data Machine Learning for Telemedicine Triage and Outbreak Surveillance ·
github.com/vaibhav7087/curago_paper_version*